MIL-T-11776E(ME)
18 May 1988
SUPERSEDING
MIL-T-11776D(ME)
15 October 1981

MILITARY SPECIFICATION

TABLE, LITHOGRAPHIC LAYOUT: TILT TOP, 24-INCH BY 30-INCH, NOMINAL ILLUMINATED AREA

This specification is approved for use by the Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a light table with an adjustable, frosted-glass working surface and an internal, fluorescent light source.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3610

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PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes Setup.
PPP-B-1055	- Barrier Material, Waterproofed Flexible.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape, Packaging, Waterproof.
PPP-T-97	- Tape, Pressure Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/6	- Plate, Identification.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-889	- Dissimilar Metals
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

2.1.2 Other Government drawings. The following other Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

ME

TA13220E9920	- Table, Lithographic Layout.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 2016 - Standard Methods of Test for Moisture Content of Wood.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The table shall be in accordance with Top Assembly Drawings TA 13220E9920 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the table. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by

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the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of preproduction models from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish tables conforming to this specification.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The table shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the table may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the table may be newly fabricated from recovered materials to the maximum extent practicable, provided the table produced meets all other requirements of this specification. Used, rebuilt, or remanufactured components, pieces and parts shall not be incorporated in the table.

3.4 Construction.

3.4.1 Wood. Wood used in the fabrication of the table shall have a moisture content of not less than 6 percent or more than 12 percent when tested in accordance with ASTM D 2016 (see 4.5.2.2).

3.4.2 Dimensions. Tolerance and gauges shall conform to the limits specified herein and on the applicable drawings.

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3.5 Ground-glass tops. The table shall be supplied with two ground-glass tabletops (working surfaces); one ground-glass tabletop shall be mounted.

3.6 Electrical components. All electrical equipment and wiring shall conform to the National Electrical Code NFPA No. 70.

3.7 Strength. When secured only by the angle brackets on each leg, the table shall support a deadweight of 500 pounds (lb) and at the same time withstand a lateral force of 100 lb applied perpendicular to the top of each of the four sides separately, without permanent deformation.

3.8 Identification marking. The table shall be identified in accordance with MIL-STD-130. The marking shall be applied to the table on plates conforming to MIL-P-15024 and MIL-P-15024/6. Plates shall be attached by screws, bolts, or rivets in a conspicuous, protected location.

3.9 Treatment and painting. The portions of the table, lithographic layout, normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704 type A, F, or G, as applicable. Color shall be the option of the contracting officer (see 6.2).

3.10 Workmanship.

3.10.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to ensure uniformity of size and shape. Precautions shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges or corners are required or where they are not detrimental to safety.

3.10.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale and other foreign matter that can be removed by chipping and wire brushing. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the table to conform to the examination and test requirements specified in section 4. Unless otherwise specified (see 6.2), the welding process used in fabrication of the table shall be the option of the contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use

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his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality assurance program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

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4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The tables shall be examined as specified herein for the following defects:

Major

101. Materials not as specified (see 3.3).
102. Material not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals, as defined in MIL-STD-889, are not effectively insulated from each other (see 3.3.1.1).
104. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.1.2).
105. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the table unit (see 3.3.2).
106. Dimensions not in accordance with drawings (see 3.4.2).
107. Ground-glass top not as specified (see 3.5).
108. Electrical components not as specified (see 3.6).
109. Workmanship not as specified (see 3.10).
110. Metal fabrication not as specified (see 3.10.1).
111. Welding not as specified (see 3.10.2).

Minor

201. Spare glass top not provided as specified (see 3.5).
202. Identification marking missing, incomplete, or not as specified (see 3.8).
203. Treatment and painting not as specified (see 3.9).
204. Color not as specified by contracting officer (see 3.9 and 6.2).

4.5.2 Tests

4.5.2.1 Strength. When secured only by the angle brackets on each leg, the table shall be uniformly loaded with a deadweight of 500 lb and while so loaded shall be subjected to a lateral force of 100 lb perpendicular to the top of each of the four sides separately. The table shall then be examined. Evidence of permanent deformation shall constitute failure of this test (see 3.7).

4.5.2.2 Moisture content. The moisture content of the wood shall be measured in accordance with ASTM D 2016. Evidence of moisture content less than 6 percent or more than 12 percent shall constitute failure of this test (see 3.4.1).

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

113.	Materials, methods, and containers not as specified for level A or C (see 5.1).
114.	Disassembly not as specified for level A (see 5.1.1.1).
115.	Unpainted surfaces of components requiring a contact preservative not coated as specified for level A (see 5.1.1.3).
116.	Openings into electrical components not sealed as specified for level A (see 5.1.1.4).
117.	Fluorescent lamps not cushioned and placed in a box as specified for level A (see 5.1.1.5).
118.	Mounted-glass working surface not cushioned, covered with barrier material, and secured with tape as specified for level A (see 5.1.1.7).
119.	Extra working glass not cushioned, placed in a box and secured as specified for level A (see 5.1.1.8).
120.	Straightedge assembly not preserved as specified for level A (see 5.1.1.9).
121.	Maintenance tools not preserved in accordance with the referenced document for level A (see 5.1.1.11).
122.	Technical publications not preserved as specified for level A (see 5.1.1.12).
123.	Consolidation not as specified for level A (see 5.1.1.13).
124.	Table not secured and protected against abrasion within the box in accordance with the referenced document for level A or C (see 5.2.1 and 5.2.2).
125.	Shipping containers not as specified for level A or C (see 5.2).
126.	Strapping not as specified (see 5.2.1).
127.	Marking missing, illegible, incorrect, or incomplete (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Disassembly. Parts that would increase cubage or that are vulnerable to damage, such as fluorescent lamps, starters, straightedge assembly with cord, and the tray and bracket shall be removed from the table. When bolts, nuts, washers and pins are removed, they shall be replaced on one of the mating parts and secured to prevent their loss.

5.1.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in, and shall be applied in accordance with, MIL-P-116.

5.1.1.3 Unpainted surfaces. Unpainted exterior metal surfaces of component parts of the table, such as angles, brackets, rods, swivels, bearings, bolts and nuts, pins and washers or any other metal surfaces which may become exposed by disassembly and which require a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative.

5.1.1.4 Electrical components. Openings in electrical components, such as starter openings, switches and receptacles and exposed ends of wire, plugs and other connections shall be sealed with tape conforming to PPP-T-60, type IV. The rubber cord shall be coiled to a minimum safe diameter and the coil shall be secured to the table with tape specified herein.

5.1.1.5 Fluorescent lamps. Each fluorescent lamp shall be wrapped with cushioning material conforming to PPP-C-843, type II, class A or B and secured with tape conforming to PPP-T-60, type IV. The lamps shall be placed together in a close-fitting box conforming to PPP-B-636, W6c.

5.1.1.6 Lamp starters. The lamp starters shall be preserved together in accordance with MIL-P-116, method III.

5.1.1.7 Glass working surface. The mounted-glass working surface shall be covered with 1-inch-thick cushioning material conforming to PPP-C-843, type II, class B, and the cushioning material shall be covered with barrier material conforming to PPP-B-1055, class C-1, secured in place with tape conforming to PPP-T-60, type IV.

5.1.1.8 Extra glass working surface. The extra glass shall be wrapped with 1-inch-thick cushioning material conforming to PPP-C-843, type II, class B, and secured in place with tape conforming to PPP-T-60, type IV. The cushioned glass shall be placed in a close-fitting box conforming to PPP-B-636, W6c, and placed flat and secured on the tabletop with tape conforming to PPP-T-97, type IV, size 3/4-inch.

5.1.1.9 Straightedge assembly. The cord shall be coiled and secured. The straightedge assembly shall be preserved in accordance with MIL-P-116,

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method IC-2, coating unprotected metal parts with type P-11 preservative. The container shall be a box conforming to PPP-B-566 or PPP-B-676.

5.1.1.10 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.1.1.11 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A preservation requirements of PPP-P-40.

5.1.1.12 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3, and secured to the tabletop with tape.

5.1.1.13 Consolidation. Preserved components and parts shall be placed and secured within the lamp housing. Cushioning material conforming to PPP-C-843, type II, class A or B shall be provided to prevent damage to the items or to the housing.

5.1.2 Level C. The table with all components and accessories shall be preserved in a manner which will prevent damage and corrosion during shipment to the first receiving activity. Particular attention shall be paid to use of cushioning to prevent damage.

5.2 Packing. Packing shall be level A or C, as specified (see 6.2).

5.2.1 Level A. Each complete table shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Finished surfaces of the table shall be protected against abrasion and the contents secured within the box in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.2.2 Level C. Each complete table shall be packed in a box conforming to PPP-B-636, class domestic, variety and grade appropriate for the size and weight of the contents. Surfaces of the table shall be protected against abrasion and scratching; cushioning shall be used to prevent movement of the contents within the box. The box shall be closed and sealed in accordance with the box specification.

5.3 Marking. Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The light table is to be used for opaquing, layout, and inspections required in preparing film negatives and other photographic and translucent material used in lithographic processes.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is not required (see 3.2).

c. Timeframe required for submission of first article (see 3.2).

d. Color required by contracting officer (see 3.9).

e. When the welding process will not be at the option of the contractor (see 3.10.2).

f. Level of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the table should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each table.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair and maintenance tools as necessary (including any special tools) and instructions regarding shipment of tables.

6.6 Subject term (key word) listing).

Layout table
Lithographic table, layout
Table lithographic

6.7 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activity:
DLA-GS

Preparing activity:
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Project 3610-A342

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-11776	Table, Lithographic Layout Tilt Top, 24-Inch By 30" Nominal Illuminated Area

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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